Random Variables and Expectation

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

Review: Outcomes, events, and probability

Outcome ω	One possible result of an experiment.
Sample space S	The set of all possible outcomes.
Event $E \subseteq S$	A collection of outcomes; E happens when $\omega \in E$.

Review: Outcomes, events, and probability

Outcome ω — One possible result of an experiment.

Sample space S — The set of all possible outcomes.

Event $E \subseteq S$ — A collection of outcomes; E happens when $\omega \in E$.

Classical model: finite, nonempty S; equally likely outcomes.

$$\Pr[E] = \frac{|E|}{|S|}.$$

Review: Outcomes, events, and probability

Outcome ω — One possible result of an experiment.

Sample space S — The set of all possible outcomes.

Event $E \subseteq S$ — A collection of outcomes; E happens when $\omega \in E$.

Classical model: finite, nonempty S; equally likely outcomes.

$$\Pr[E] = \frac{|E|}{|S|}.$$

General discrete model: add the probabilities in E.

$$\Pr[E] = \sum_{\omega \in E} \Pr[\{\omega\}].$$

Review: Probability rules

Conditional probability $\Pr[E \mid F] = \dfrac{\Pr[E \text{ and } F]}{\Pr[F]}$

Multiplication rule $\Pr[E \text{ and } F] = \Pr[F]\Pr[E \mid F]$

Independence (definition) E, F are independent $\iff \Pr[E \text{ and } F] = \Pr[E]\Pr[F]$

Bayes' rule $\Pr[E \mid F] = \dfrac{\Pr[F \mid E]\Pr[E]}{\Pr[F]}$

Conditioning events must have positive probability.

Review: Probability rules

Conditional probability $\Pr[E \mid F] = \dfrac{\Pr[E \text{ and } F]}{\Pr[F]}$

Multiplication rule $\Pr[E \text{ and } F] = \Pr[F]\Pr[E \mid F]$

Independence (definition) E, F are independent $\iff \Pr[E \text{ and } F] = \Pr[E]\Pr[F]$

Bayes' rule $\Pr[E \mid F] = \dfrac{\Pr[F \mid E]\Pr[E]}{\Pr[F]}$

Conditioning events must have positive probability.

These rules still apply to events defined by a random variable.

Example: Three coin flips

Flip a fair coin three times independently.

$$S = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\}.$$

What is the probability of exactly two heads?

Example: Three coin flips

Flip a fair coin three times independently.

$$S = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\}.$$

What is the probability of exactly two heads?

$$E = \{HHT, HTH, THH\}, \qquad \Pr[E] = \frac{3}{8}.$$

Example: Three coin flips

Flip a fair coin three times independently.

$$S = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\}.$$

What is the probability of exactly two heads?

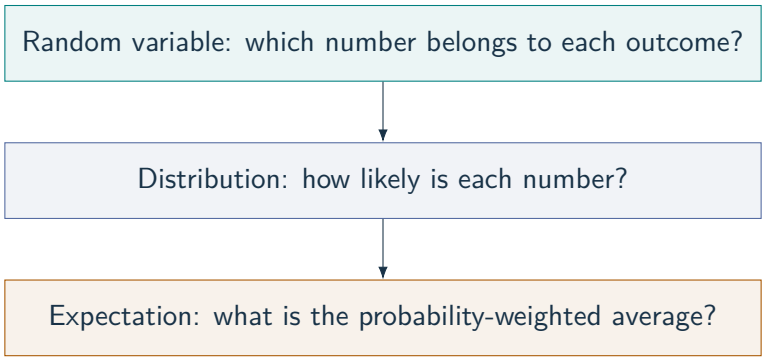
$$E = \{HHT, HTH, THH\}, \qquad \Pr[E] = \frac{3}{8}.$$

Let X record the **number of heads**:

$$X(HHT) = X(HTH) = X(THH) = 2.$$

The same event is now written $\{X = 2\}$, so $\Pr[X = 2] = 3/8$.

From outcomes to averages



SECTION 1

Random Variables

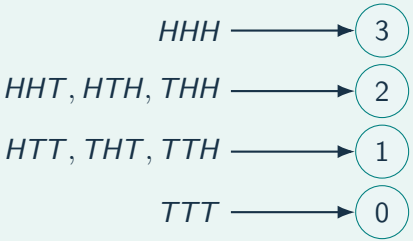
Random variables

MUST UNDERSTAND

Definition: A random variable is a function $X : S \to \mathbb{R}$, where S is the sample space. It assigns one real number to each outcome.

Example: Three coin flips

Let X be the number of heads.



Examples: Random variables

Random experiment	Number recorded
Throw two dice.	$X =$ sum of the two results.

Examples: Random variables

Random experiment	Number recorded
Throw two dice.	$X =$ sum of the two results.
Draw 5 cards from a standard 52-card deck without replacement.	$Y =$ number of aces drawn.

Examples: Random variables

Random experiment	Number recorded
Throw two dice.	X = sum of the two results.
Draw 5 cards from a standard 52-card deck without replacement.	Y = number of aces drawn.
Roll a fair die independently until the first 6.	T = number of rolls, including the final 6.

Examples: Random variables

Random experiment	Number recorded
Throw two dice.	$X =$ sum of the two results.
Draw 5 cards from a standard 52-card deck without replacement.	$Y =$ number of aces drawn.
Roll a fair die independently until the first 6.	$T =$ number of rolls, including the final 6.
Draw one card from a standard 52-card deck.	$I = 1$ if it is a heart; $I = 0$ otherwise.

Each outcome determines one numerical value.

Calculations with random variables

For any real-valued function g defined on the possible values of X,

$$\boldsymbol{g(X)} \text{ is also a random variable.}$$

Examples: X^2 and e^X.

The same holds for $h(X, Y)$, such as $X + Y$, when X, Y share a sample space.

SECTION 2

Distributions

Example: Three coin tosses

Toss a fair coin three times independently. Let X count heads.

Each of the eight outcomes has probability $1/8$.

$X = 0$	$X = 1$	$X = 2$	$X = 3$
TTT	HTT THT TTH	HHT HTH THH	HHH

Example: Three coin tosses

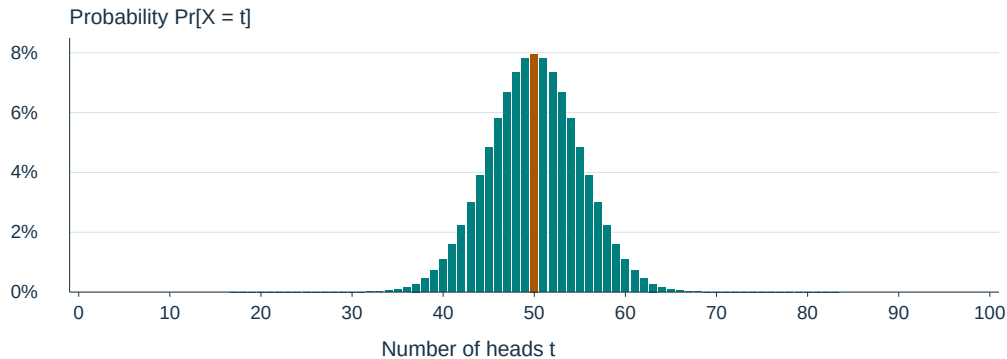
Toss a fair coin three times independently. Let X count heads.

Each of the eight outcomes has probability $1/8$.

$X = 0$	$X = 1$	$X = 2$	$X = 3$
TTT	HTT THT TTH	HHT HTH THH	HHH
$\Pr[X = 0] = \frac{1}{8}$	$\Pr[X = 1] = \frac{3}{8}$	$\Pr[X = 2] = \frac{3}{8}$	$\Pr[X = 3] = \frac{1}{8}$

Example: 100 coin tosses

Toss a fair coin 100 times independently. Let X count heads.



Each bar shows $\Pr[X = t]$: the probability of exactly t heads.

Distribution

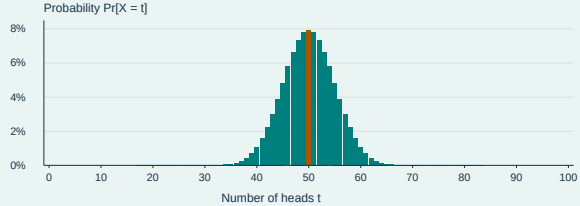
MUST UNDERSTAND

Definition: For discrete X, its distribution specifies $\Pr[X = t]$ for each value t.

$$\Pr[X = t] = \sum_{\substack{\omega \in S \\ X(\omega) = t}} \Pr[\{\omega\}].$$

Add the probabilities of all outcomes that make $X = t$.

Example: 100 independent coin tosses



Properties of a discrete distribution

Let V be the set of possible values of X.

1. Probabilities lie between 0 and 1. $0 \leq \Pr[X = t] \leq 1.$

Properties of a discrete distribution

Let V be the set of possible values of X.

1. **Probabilities lie between 0 and 1.** $0 \leq \Pr[X = t] \leq 1.$
2. **All probabilities sum to 1.** $\sum_{t \in V} \Pr[X = t] = 1.$

Properties of a discrete distribution

Let V be the set of possible values of X.

1. **Probabilities lie between 0 and 1.** $0 \leq \Pr[X = t] \leq 1.$
2. **All probabilities sum to 1.** $\sum_{t \in V} \Pr[X = t] = 1.$
3. **For several values, add their probabilities.**

$$\Pr[X \in A] = \sum_{t \in A} \Pr[X = t], \qquad A \subseteq V.$$

Properties of a discrete distribution

Let V be the set of possible values of X.

1. **Probabilities lie between 0 and 1.** $0 \leq \Pr[X = t] \leq 1.$
2. **All probabilities sum to 1.** $\sum_{t \in V} \Pr[X = t] = 1.$
3. **For several values, add their probabilities.**

$$\Pr[X \in A] = \sum_{t \in A} \Pr[X = t], \qquad A \subseteq V.$$

Example: For three independent fair tosses, X counts heads:

$$\Pr[X \geq 2] = \Pr[X = 2] + \Pr[X = 3] = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}.$$

Independent random variables

Definition: Discrete random variables X, Y are independent if and only if

$$\Pr[X = x \text{ and } Y = y] = \Pr[X = x] \Pr[Y = y]$$

for **every pair of values** x, y.

Independent random variables

Definition: Discrete random variables X, Y are independent if and only if

$$\Pr[X = x \text{ and } Y = y] = \Pr[X = x] \Pr[Y = y]$$

for **every pair of values** x, y.

Interpretation: Learning the value of one does not change the distribution of the other.

Independent random variables

Definition: Discrete random variables X, Y are independent if and only if

$$\Pr[X = x \text{ and } Y = y] = \Pr[X = x] \Pr[Y = y]$$

for **every pair of values** x, y.

Interpretation: Learning the value of one does not change the distribution of the other.

$$\Pr[X = x \mid Y = y] = \Pr[X = x] \qquad \text{when } \Pr[Y = y] > 0.$$

Example: Two dice

Roll two fair dice independently.

X — First result

Y — Second result

$Z = X + Y$ — Sum

X **and** Y

X **and** Z

Example: Two dice

Roll two fair dice independently.

X — First result

Y — Second result

$Z = X + Y$ — Sum

X and Y: independent **X and Z**

For every $x, y \in \{1, \ldots, 6\}$,

$$\Pr[X = x \text{ and } Y = y] = \frac{1}{36}$$

$$= \tfrac{1}{6} \cdot \tfrac{1}{6} = \Pr[X = x] \Pr[Y = y].$$

Example: Two dice

Roll two fair dice independently.

X — First result

Y — Second result

$Z = X + Y$ — Sum

X and Y: independent

For every $x, y \in \{1, \ldots, 6\}$,

$$\Pr[X = x \text{ and } Y = y] = \frac{1}{36}$$

$$= \tfrac{1}{6} \cdot \tfrac{1}{6} = \Pr[X = x] \Pr[Y = y].$$

X and Z: not independent

If $Z = 12$, only $(6, 6)$ is possible:

$$\underbrace{\Pr[X = 6 \mid Z = 12]}_{1} \neq \underbrace{\Pr[X = 6]}_{1/6}.$$

Knowing the sum changes the first result's distribution.

Classical Discrete Distribution

Distribution Name	Description	Example
Uniform	n outcomes, each with probability $1/n$.	A fair die result.

Classical Discrete Distribution

Distribution Name	Description	Example
Uniform	n outcomes, each with probability $1/n$.	A fair die result.
Bernoulli	Success (1) with probability p; otherwise 0.	One coin toss: heads 1, tails 0.

Classical Discrete Distribution

Distribution Name	Description	Example
Uniform	n outcomes, each with probability $1/n$.	A fair die result.
Bernoulli	Success (1) with probability p; otherwise 0.	One coin toss: heads 1, tails 0.
Binomial	Count successes in n trials.	Count heads in 100 tosses.

Binomial/geometric: independent trials with fixed success probability p ($p > 0$ for geometric).

Classical Discrete Distribution

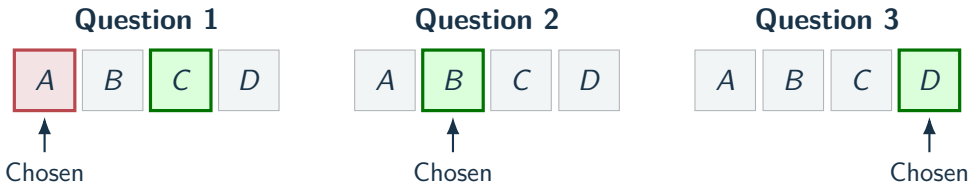
Distribution Name	Description	Example
Uniform	n outcomes, each with probability $1/n$.	A fair die result.
Bernoulli	Success (1) with probability p; otherwise 0.	One coin toss: heads 1, tails 0.
Binomial	Count successes in n trials.	Count heads in 100 tosses.
Geometric	Count trials up to and including the first success.	Count rolls until the first 6.

Binomial/geometric: independent trials with fixed success probability p ($p > 0$ for geometric).

Example: Guessing on a quiz

A quiz has 3 questions, each with 4 options and exactly one correct answer.

A student guesses independently, choosing each option with probability 1/4.



Let X be the number of correct answers.

What is the distribution of X?

Quiz: The probability of one pattern

Consider $X = 2$: exactly two correct answers.

For each question: $\Pr[C] = 1/4$, $\Pr[W] = 3/4$.

C = correct, W = wrong.

Quiz: The probability of one pattern

Consider $X = 2$: exactly two correct answers.

For each question: $\Pr[C] = 1/4$, $\Pr[W] = 3/4$.

C = correct, W = wrong.

Consider the pattern: correct, correct, wrong.

C		C		W
$1/4$	$\times$	$1/4$	$\times$	$3/4$

Independent guesses: multiply the probabilities.

$$\Pr[\text{CCW}] = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{64}.$$

Quiz: Count all three patterns

Choose which 2 of the 3 questions are correct: $\binom{3}{2} = 3$ choices.

Q1	Q2	Q3
C	C	W
C	W	C
W	C	C

Quiz: Count all three patterns

Choose which 2 of the 3 questions are correct: $\binom{3}{2} = 3$ choices.

Q1	Q2	Q3	
C	C	W	$\Pr[\text{CCW}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$
C	W	C	$\Pr[\text{CWC}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$
W	C	C	$\Pr[\text{WCC}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$

Quiz: Count all three patterns

Choose which 2 of the 3 questions are correct: $\binom{3}{2} = 3$ choices.

Q1	Q2	Q3	
C	C	W	$\Pr[\text{CCW}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$
C	W	C	$\Pr[\text{CWC}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$
W	C	C	$\Pr[\text{WCC}] = \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{3}{64}$

The patterns are disjoint, so add their probabilities.

$$\Pr[X = 2] = \binom{3}{2} \left(\frac{1}{4}\right)^2 \frac{3}{4} = \frac{9}{64}.$$

Binomial distribution

Consider n independent trials. Each gives a binary result:

$$\text{Yes with probability } p, \qquad \text{No with probability } 1 - p.$$

$0 \leq p \leq 1$. For $p = 0$ or 1, a factor with exponent zero is 1.

Binomial distribution

Consider n independent trials. Each gives a binary result:

$$\text{Yes with probability } p, \qquad \text{No with probability } 1 - p.$$

Let X be the **total number of Yes results**.

X has a **binomial distribution**, written

$$X \sim \text{Bin}(n, p).$$

$0 \leq p \leq 1$. For $p = 0$ or 1, a factor with exponent zero is 1.

Binomial distribution

Consider n independent trials. Each gives a binary result:

$$\text{Yes with probability } p, \qquad \text{No with probability } 1-p.$$

Let X be the **total number of Yes results**.

X has a **binomial distribution**, written

$$X \sim \text{Bin}(n, p).$$

$$\Pr[X = k] = \underbrace{\binom{n}{k}}_{\#\text{cases with } k \text{ ``Yes''}} \underbrace{p^k(1-p)^{n-k}}_{\text{prob. of one outcome w/ } k \text{ ``Yes''}}, \qquad k = 0, \ldots, n.$$

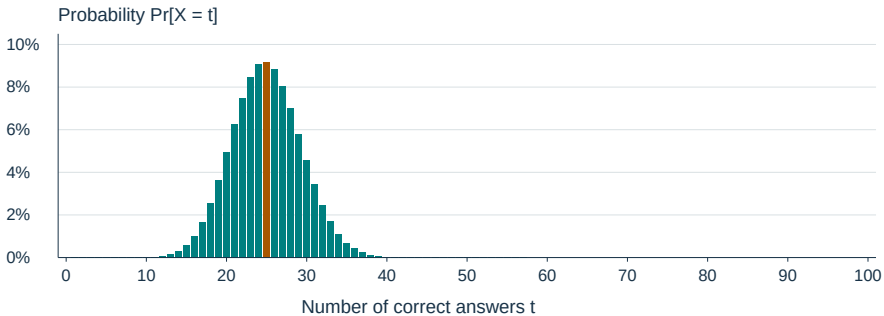
$0 \le p \le 1$. For $p = 0$ or 1, a factor with exponent zero is 1.

Example: A 100-question quiz

Guess independently on 100 questions, each with 4 equally likely options.

One correct option per question. Let X count correct answers.

$$X \sim \text{Bin}(100, 1/4).$$



SECTION 3

Expectation

Example: A two-outcome game

A game has two outcomes:

1. Win \$20 with probability $1/4$.
2. Lose \$5 with probability $3/4$.

Example: A two-outcome game

A game has two outcomes:

1. Win \$20 with probability 1/4.
2. Lose \$5 with probability 3/4.

Should you play this game many times?

Two-outcome game: The long-run average intuition

Play the game independently n times, for large n.

Outcome	Approximate count	Total gain
Win \$20	$\frac{1}{4}n$	$\frac{1}{4}n \cdot 20$
Lose \$5	$\frac{3}{4}n$	$\frac{3}{4}n \cdot (-5)$

Two-outcome game: The long-run average intuition

Play the game independently n times, for large n.

Outcome	Approximate count	Total gain
Win \$20	$\frac{1}{4}n$	$\frac{1}{4}n \cdot 20$
Lose \$5	$\frac{3}{4}n$	$\frac{3}{4}n \cdot (-5)$

$$\text{Average gain per game} \approx \frac{\frac{1}{4}n \cdot 20 + \frac{3}{4}n \cdot (-5)}{n} = \frac{1}{4} \cdot 20 + \frac{3}{4} \cdot (-5) = 1.25.$$

Two-outcome game: The long-run average intuition

Play the game independently n times, for large n.

Outcome	Approximate count	Total gain
Win \$20	$\frac{1}{4}n$	$\frac{1}{4}n \cdot 20$
Lose \$5	$\frac{3}{4}n$	$\frac{3}{4}n \cdot (-5)$

$$\text{Average gain per game} \approx \frac{\frac{1}{4}n \cdot 20 + \frac{3}{4}n \cdot (-5)}{n} = \frac{1}{4} \cdot 20 + \frac{3}{4} \cdot (-5) = 1.25.$$

The pattern: value $\times$ probability, then add.

Expectation

MUST UNDERSTAND

Denote the expectation of a random variable X by $\mathbb{E}[X]$.

The definition is:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x].$$

Expectation

MUST UNDERSTAND

Denote the expectation of a random variable X by $\mathbb{E}[X]$.

The definition is:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x].$$

$$\underbrace{\text{value}}_{x} \times \underbrace{\text{probability}}_{\Pr[X=x]} \quad \text{then add over all values.}$$

Expectation: The long-run average

Expectation describes the average over many independent repetitions.

Take independent samples $X_1, X_2, \ldots, X_n$ from the same distribution D as X.

Expectation: The long-run average

Expectation describes the average over many independent repetitions.

Take independent samples $X_1, X_2, \ldots, X_n$ from the same distribution D as X.

$$\underbrace{\frac{X_1 + X_2 + \cdots + X_n}{n}}_{\text{average of the observed values}} \approx \underbrace{\mathbb{E}[X]}_{\text{expectation}} .$$

For large n, this average is likely to be **close** to $\mathbb{E}[X]$.

Expectation: The long-run average

Expectation describes the average over many independent repetitions.

Take independent samples $X_1, X_2, \ldots, X_n$ from the same distribution D as X.

$$\underbrace{\frac{X_1 + X_2 + \cdots + X_n}{n}}_{\text{average of the observed values}} \approx \underbrace{\mathbb{E}[X]}_{\text{expectation}} .$$

For large n, this average is likely to be **close** to $\mathbb{E}[X]$.

This is the intuition behind the **law of large numbers**.

It is not a prediction of the next individual result.

Two-outcome game: Expected gain

Let X be your net gain, in dollars.

Outcome	Gain x	Probability
Win \$20	20	$1/4$
Lose \$5	-5	$3/4$

Two-outcome game: Expected gain

Let X be your net gain, in dollars.

Outcome	Gain x	Probability	Value $\times$ probability
Win \$20	20	$1/4$	$20 \cdot \frac{1}{4} = 5$
Lose \$5	-5	$3/4$	$(-5) \cdot \frac{3}{4} = -3.75$

Two-outcome game: Expected gain

Let X be your net gain, in dollars.

Outcome	Gain x	Probability	Value $\times$ probability
Win \$20	20	$1/4$	$20 \cdot \frac{1}{4} = 5$
Lose \$5	-5	$3/4$	$(-5) \cdot \frac{3}{4} = -3.75$

$$\mathbb{E}[X] = 5 - 3.75 = 1.25.$$

The expected net gain is \$1.25 per play.

Functions of a random variable

To find $\mathbb{E}[g(X)]$, transform each value before averaging:

$$\mathbb{E}[g(X)] = \sum_x g(x) \Pr[X = x].$$

We can also combine multiple variables: $g(X, Y) = X + Y$ or XY.

Functions of a random variable

To find $\mathbb{E}[g(X)]$, transform each value before averaging:

$$\mathbb{E}[g(X)] = \sum_x g(x) \Pr[X = x].$$

We can also combine multiple variables: $g(X, Y) = X + Y$ or XY.

Warning: In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.
$\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$.

Functions of a random variable

To find $\mathbb{E}[g(X)]$, transform each value before averaging:

$$\mathbb{E}[g(X)] = \sum_x g(x)\Pr[X = x].$$

We can also combine multiple variables: $g(X, Y) = X + Y$ or XY.

Warning: In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.
$\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$.

Example: The two-outcome game, $g(x) = x^2$**.**

$X = 20$ with probability $1/4$; $X = -5$ with probability $3/4$.

$$\mathbb{E}[X^2] = 20^2 \cdot \frac{1}{4} + (-5)^2 \cdot \frac{3}{4} = 118.75,$$
$$(\mathbb{E}[X])^2 = (1.25)^2 = 1.5625.$$

Linearity of expectation

MUST UNDERSTAND

For finite expectations and constants a, b:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Linearity of expectation

MUST UNDERSTAND

For finite expectations and constants a, b:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Extremely powerful: it works regardless of whether X and Y are independent.

Why linearity holds?

For a finite sample space, group outcomes by their value:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x] = \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}].$$

Use the same original outcome ω for both variables:

$$\mathbb{E}[X + Y] = \sum_{\omega \in S} (X(\omega) + Y(\omega)) \Pr[\{\omega\}]$$

Why linearity holds?

For a finite sample space, group outcomes by their value:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x] = \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}].$$

Use the same original outcome ω for both variables:

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_{\omega \in S} (X(\omega) + Y(\omega)) \Pr[\{\omega\}] \\ &= \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}] + \sum_{\omega \in S} Y(\omega) \Pr[\{\omega\}]\end{aligned}$$

Why linearity holds?

For a finite sample space, group outcomes by their value:

$$\mathbb{E}[X] = \sum_x x \Pr[X = x] = \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}].$$

Use the same original outcome ω for both variables:

$$\begin{aligned}\mathbb{E}[X + Y] &= \sum_{\omega \in S} (X(\omega) + Y(\omega)) \Pr[\{\omega\}] \\ &= \sum_{\omega \in S} X(\omega) \Pr[\{\omega\}] + \sum_{\omega \in S} Y(\omega) \Pr[\{\omega\}] \\ &= \mathbb{E}[X] + \mathbb{E}[Y].\end{aligned}$$

Exercise: Prove $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$.

Expectation of Binomial Distribution

Recall: n independent trials; Yes with probability p, No with probability $1 - p$.

$X =$ total number of Yes results, so $X \sim \text{Bin}(n, p)$.

$$\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

Expectation of Binomial Distribution

Recall: n independent trials; Yes with probability p, No with probability $1 - p$.

$X =$ total number of Yes results, so $X \sim \text{Bin}(n, p)$.

$$\Pr[X = k] = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

Using the definition of expectation:

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \Pr[X = k] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}.$$

Can we find the expectation without evaluating this sum?

Expectation of Binomial Distribution

A simpler calculation: linearity of expectation.

Define

$$I_j = \begin{cases} 1, & \text{the } j\text{-th trial is Yes,} \\ 0, & \text{the } j\text{-th trial is No.} \end{cases}$$

Expectation of Binomial Distribution

A simpler calculation: linearity of expectation.

Define

$$I_j = \begin{cases} 1, & \text{the } j\text{-th trial is Yes,} \\ 0, & \text{the } j\text{-th trial is No.} \end{cases}$$

$$\mathbb{E}[I_j] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

The total number of Yes results is $X = I_1 + \cdots + I_n$.

Expectation of Binomial Distribution

A simpler calculation: linearity of expectation.

Define

$$I_j = \begin{cases} 1, & \text{the } j\text{-th trial is Yes,} \\ 0, & \text{the } j\text{-th trial is No.} \end{cases}$$

$$\mathbb{E}[I_j] = 1 \cdot p + 0 \cdot (1 - p) = p.$$

The total number of Yes results is $X = I_1 + \cdots + I_n$.

$$\mathbb{E}[X] = \mathbb{E}[I_1] + \cdots + \mathbb{E}[I_n] = p + \cdots + p = np.$$

Expectation of Binomial Distribution

$$X \sim \mathrm{Bin}(n, p) \quad \Longrightarrow \quad \mathbb{E}[X] = np.$$

Counting with indicators

In general, an indicator I_A records whether an event A happens:

$$I_A = \begin{cases} 1, & A \text{ happens,} \\ 0, & A \text{ does not happen.} \end{cases}$$

Counting with indicators

In general, an indicator I_A records whether an event A happens:

$$I_A = \begin{cases} 1, & A \text{ happens,} \\ 0, & A \text{ does not happen.} \end{cases}$$

$$\mathbb{E}[I_A] = 1 \Pr[A] + 0 \Pr[\overline{A}] = \Pr[A].$$

Counting with indicators

In general, an indicator I_A records whether an event A happens:

$$I_A = \begin{cases} 1, & A \text{ happens,} \\ 0, & A \text{ does not happen.} \end{cases}$$

$$\mathbb{E}[I_A] = 1 \Pr[A] + 0 \Pr[\overline{A}] = \Pr[A].$$

If X counts how many events happen,

$$X = I_{A_1} + \cdots + I_{A_n}.$$

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[I_{A_1}] + \cdots + \mathbb{E}[I_{A_n}] \\ &= \Pr[A_1] + \cdots + \Pr[A_n] = \sum_{i=1}^{n} \Pr[A_i]. \end{aligned}$$

Example: A random permutation

Uniformly permute $1, 2, \ldots, n$.

How many numbers stay in their original positions in expectation?



This example has two correct positions.

Random permutation: Solution

Let $I_i = 1$ when position i contains i, and 0 otherwise.

$$\Pr[I_i = 1] = \frac{1}{n}.$$

Random permutation: Solution

Let $I_i = 1$ when position i contains i, and 0 otherwise.

$$\Pr[I_i = 1] = \frac{1}{n}.$$

$$X = \sum_{i=1}^{n} I_i, \qquad \mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = n\frac{1}{n} = 1.$$

Random permutation: Solution

Let $I_i = 1$ when position i contains i, and 0 otherwise.

$$\Pr[I_i = 1] = \frac{1}{n}.$$

$$X = \sum_{i=1}^{n} I_i, \qquad \mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = n\frac{1}{n} = 1.$$

The indicators need not be independent.

Summary

Random variable	$X : S \to \mathbb{R}$
Distribution	$\Pr[X = x]$ for each value x
Expectation	$\mathbb{E}[X] = \sum_x x \Pr[X = x]$
Linearity	$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
Counting	$\mathbb{E}[\sum_i I_{A_i}] = \sum_i \Pr[A_i]$

Optional: Waiting for a success

Independent trials; each succeeds with probability $0 < p \leq 1$. Let G be the trial number of the first success.



Optional: Waiting for a success

Independent trials; each succeeds with probability $0 < p \leq 1$. Let G be the trial number of the first success.



$$\Pr[G = k] = (1 - p)^{k-1} p, \qquad k \geq 1.$$

Optional: Waiting for a success

Independent trials; each succeeds with probability $0 < p \leq 1$. Let G be the trial number of the first success.



$$\Pr[G = k] = (1-p)^{k-1}p, \qquad k \geq 1.$$

A wait of G trials contributes one for each $j \leq G$.

$$\mathbb{E}[G] = \sum_{j=1}^{\infty} \Pr[G \geq j] = \sum_{j=1}^{\infty} (1-p)^{j-1} = \frac{1}{p}.$$

Optional: Several successes

Let W_r count trials until the rth success.

$$W_r = \underbrace{G_1 + \cdots + G_r}_{\text{successive waits}}.$$

Optional: Several successes

Let W_r count trials until the rth success.

$$W_r = \underbrace{G_1 + \cdots + G_r}_{\text{successive waits}}.$$

$$\mathbb{E}[W_r] = \frac{r}{p}.$$

Optional: Several successes

Let W_r count trials until the rth success.

$$W_r = \underbrace{G_1 + \cdots + G_r}_{\text{successive waits}}.$$

$$\mathbb{E}[W_r] = \frac{r}{p}.$$

A die result is accepted unless it is 6:

$$\mathbb{E}[W_{10}] = \frac{10}{5/6} = 12 \text{ rolls.}$$

Optional: Average-case search

A target occurs in a list of n distinct elements. Its position is uniformly distributed.

$$\Pr[T = i] = \frac{1}{n}, \qquad i = 1, \ldots, n,$$

where T is the number of comparisons in linear search.

Optional: Average-case search

A target occurs in a list of n distinct elements. Its position is uniformly distributed.

$$\Pr[T = i] = \frac{1}{n}, \qquad i = 1, \ldots, n,$$

where T is the number of comparisons in linear search.

$$\mathbb{E}[T] = \frac{1 + 2 + \cdots + n}{n} = \frac{n+1}{2}.$$

Problem set

1. Two independent fair flips: find the head-count distribution, $\mathbb{E}[X]$, and $\mathbb{E}[X(X+1)]$.
2. Let X count successes in five independent trials with $p = 0.4$. Find $\Pr[X = 2]$ and $\mathbb{E}[X]$.
3. Draw five cards uniformly without replacement from a standard 52-card deck. Find the expected number of aces.
4. For a random permutation of six numbers, find the expected correct positions. Are the indicators independent?

Full solutions and optional waiting-time/search exercises are in the note.